

UNIX Non-kernel based AX.25 protocol stack
examples using DL9SAU's enhanced libax25

<https://github.com/unix-ax25-stack>

2026-10-04

Background:

Starting with the Linux 7.1.x kernel series, the following HAMRADIO kernel modules were removed:

```
CONFIG_HAMRADIO
CONFIG_AX25
CONFIG_AX25_DAMA_SLAVE
CONFIG_NETROM
CONFIG_ROSE
CONFIG_MKISS
CONFIG_6PACK
CONFIG_BPQETHER
CONFIG_BAYCOM_SER_FDX
CONFIG_BAYCOM_SER_HDX
CONFIG_BAYCOM_PAR
CONFIG_BAYCOM_EPP
CONFIG_YAM
CONFIG_LEGACY_PTYS
CONFIG_LEGACY_PTY_COUNT
```

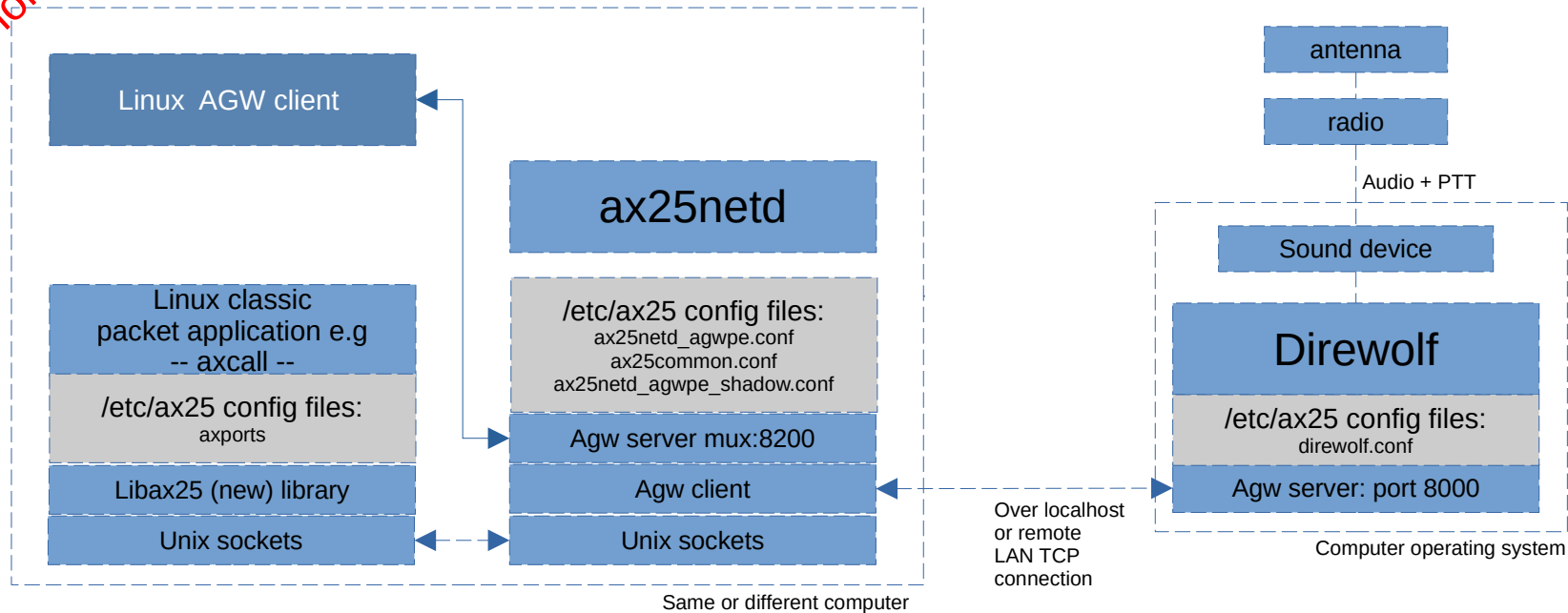
This code removal effectively orphaned dozens of AX.25 packet radio programs which the global amateur community continues to depend on. This new project is an effort to provide an alternative libax25 library and suite of tools to give these important packet radio programs a future using a “user land” AX.25 protocol stack.

Scenario #1: Basic Packet Client configuration

Direwolf w/ sound device with AGW server to ax25netd to Linux native AX.25 client application

- Single AX.25 stack

Scenario: priority #1



Scenario #1: Basic Packet Client configuration FAQ

Q: Why should I use ax25netd's AGW port 8200 vs. Direwolf's AGW port 8000?

A: The two AGW interfaces are different. If you want all applications traffic be accounted for, mheard details be up to date, to see eachother's traffic, etc., Use the ax25netd port 8100 for all AGW client applications

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Q: What is the /etc/ax25/ax25netd_agwpe_shadow.conf file for

A: This file is optional and only needed if you use an AGW server that supports authentication. Direwolf v1.9.x as of 10/2/26 does not support authentication

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Q: What is /etc/ax25/wampes.conf used for?

A: ??

Scenario #1: Basic Packet Client configuration

- Either keep “radio0” as your AX.25 interface alias name or change it in the axports and agwpe.conf files
- Change all “n0call” lines to reflect your callsign and chosen SSID (no specified SSID defaults to mean “-0”)

/etc/ax25/axports

```
radio0 n0call 255 4 Direwolf channel 0
```

/etc/ax25/ax25netd_agwpe.conf

```
radio0    localhost 8000
loop      -        -    # local loopback upstream
```

/etc/ax25/ax25common.conf

```
#loop tcp 8200
#loop socket /var/run/ax25/sockets/ax25netd.sock
#loop group hams
```

/etc/ax25/direwolf.conf

```
#User chosen device: Generic USB Audio Device
ADEVICE plughw:1,0
CHANNEL 0
MYCALL N0CALL
MODEM 1200
PTT /dev/ttyUSB0 RTS
#Setting of 25 means 250ms
TXDELAY 25
#Setting of 5 means 5ms
TXTAIL 5
AGWPORT 8000
FIX_BITS 0 AX25
```

Download URL for recommended scenario config tarball on Github

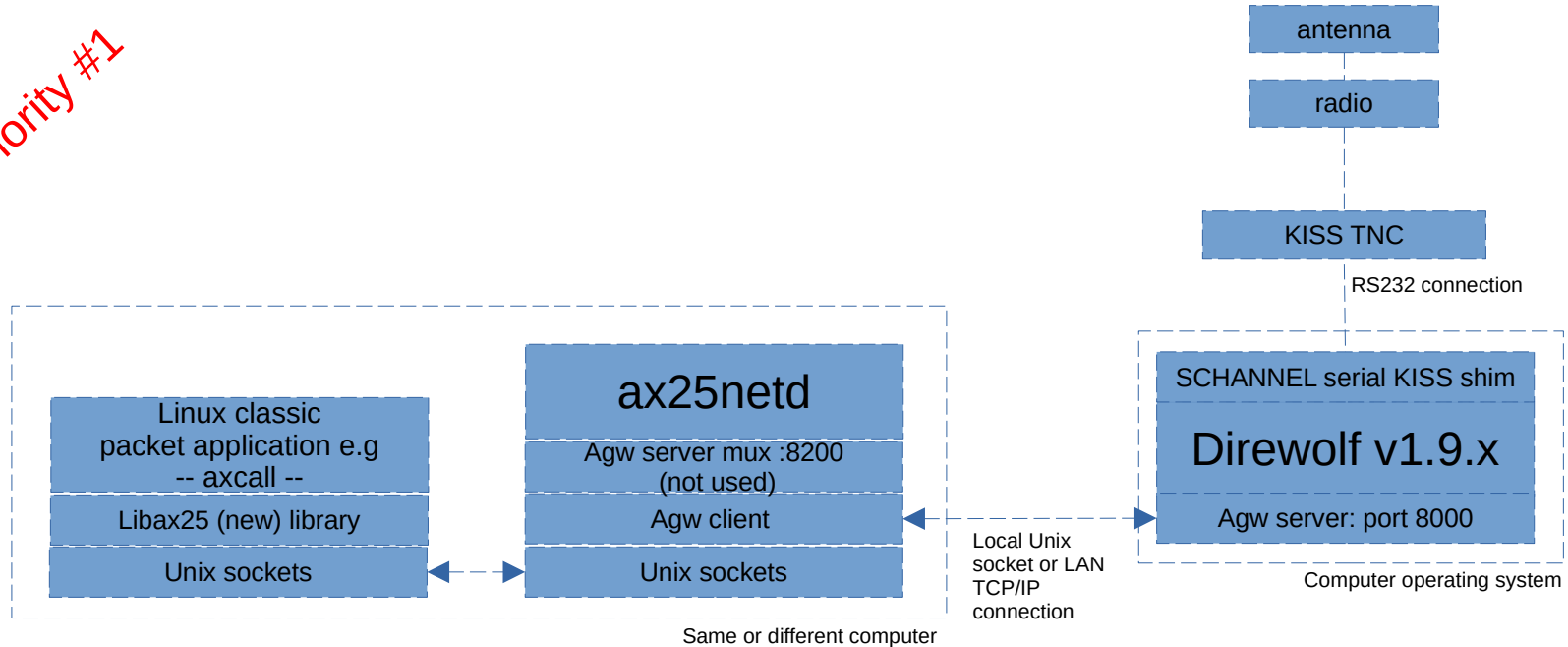
- The files in the tarball should include all standard file comments, etc. to be very complete

Scenario #2: Basic Packet Client configuration:

Direwolf v1.9.x using hardware KISS TNC (no sound device) with AGW server to ax25netd to Linux native AX.25 client application

- Single AX.25 stack

Scenario: priority #1

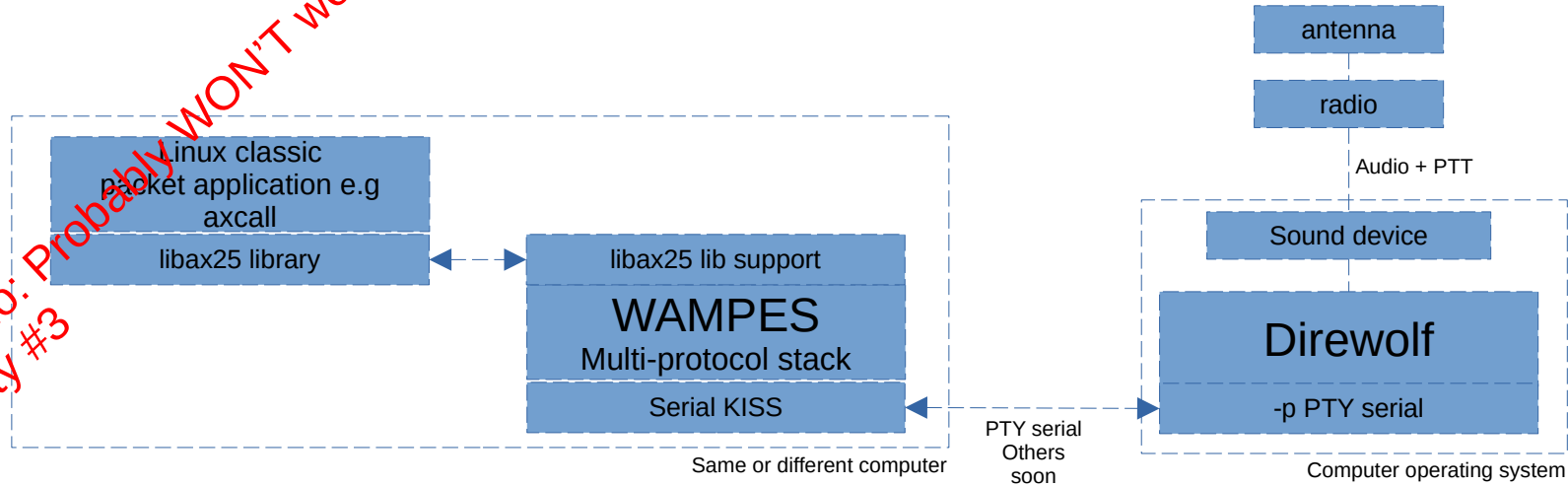


See <https://groups.io/g/direwolf/message/11599> for SCHANNEL and NCHANNEL Direwolf examples

Scenario #3: Medium Complexity Packet server configuration

Direwolf+AGW to WAMPES to Uronode node server via AX.25 and Netrom

- Dual AX.25 stacks



- Won't WAMPES give us netrom support?
Maybe it's the interconnect to ax25netd and it not supporting netrom yet is the issue

WAMPES supports:

libax25 unix sockets

serial kiss

serial 6pack

axip

axudp

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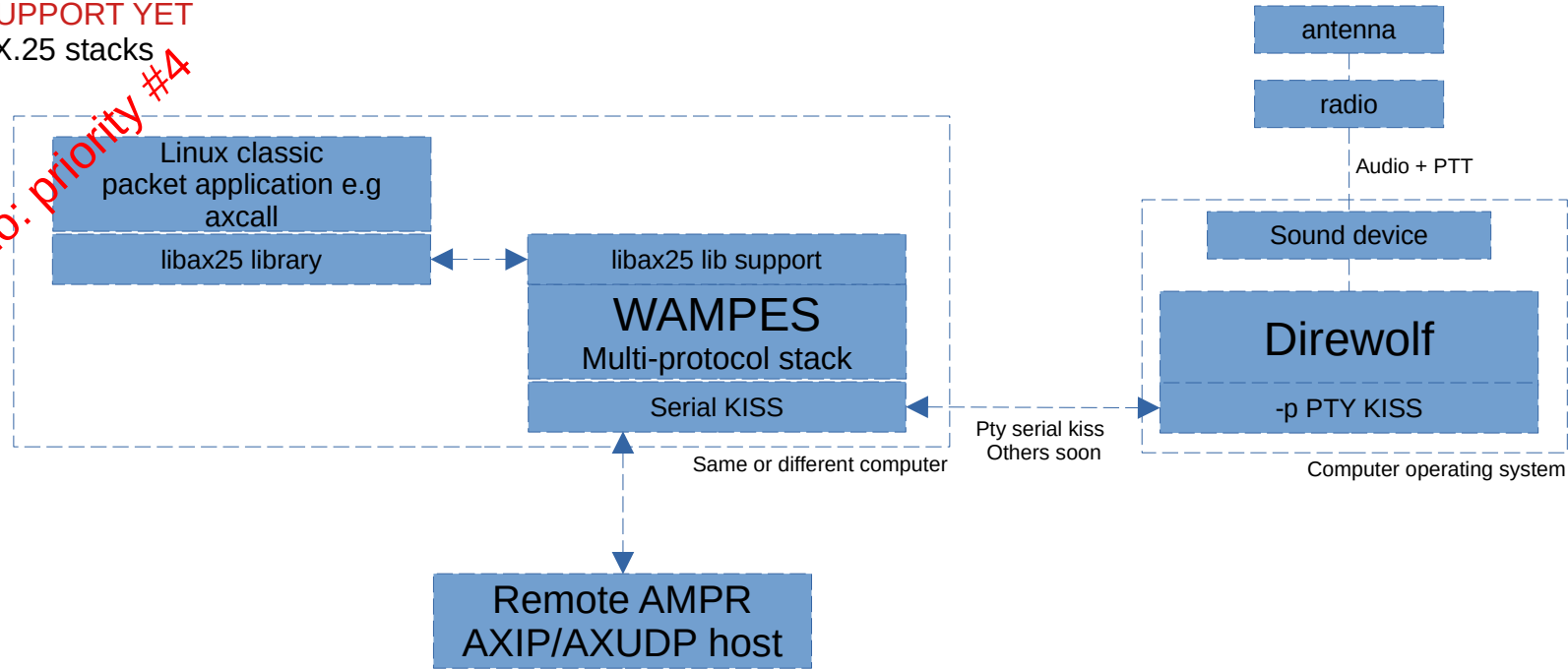
tcp-kiss is on the todo list

axtcp is on the todo list

Scenario #4: Direwolf+AGW to WAMPES to Linux native AX.25 client + AMPR AXUDP peer(s)

- NOT SUPPORT YET
- Dual AX.25 stacks

Scenario: priority #4



Questions:

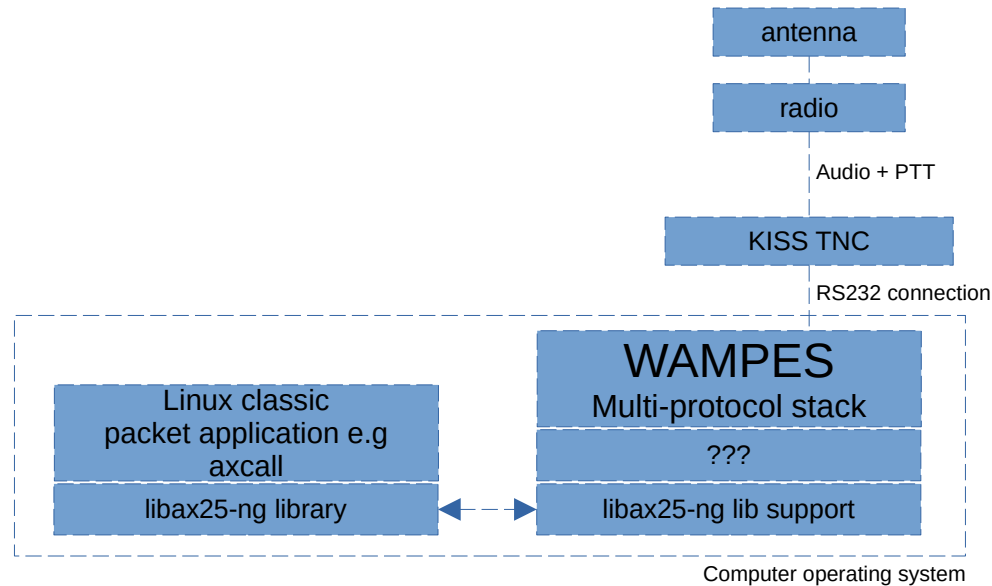
- NOT supported yet -- what isn't supported? AXIP AXTCP?

Scenario #X:

Direwolf v1.9.x using hardware KISS TNC (no sound device) with AGW server to wampes to ax25netd to Linux native AX.25 client application

- Single AX.25 stack

Scenario: priority #2



Comment:

- Serial-KISS and TCP-KISS are a weaker protocol interface than AGW : AVOID if possible

- Other scenarios to consider?

• Tested libax25 Linux application testing with new solution

Program	Status	Notes
Direwolf	Works!	
Qtsoundmodem	Not tested yet	
GreyWolf		
Axcall, beacon, ax25d, listen / etc https://github.com/unix-ax25-stack	Works	See dedicated slide on all test results
Converse-saupp https://github.com/dl9sau/conversd-saupp	Not tested yet	
PAT https://github.com/la5nta/pat	Not tested yet	
Linpac https://sourceforge.net/projects/linpac/	Under test	Digipeating?
FBB and FPAC https://sourceforge.net/projects/linfbb/	Not tested yet	No ROSE support (maybe future?)
UROnode https://sourceforge.net/projects/uronode/	Not tested yet	NETROM support?
LinBPQ https://www.cantab.net/users/john.wiseman/Documents/LinBPQGuides.html	Not tested yet	
JNOS https://www.langelaar.net/	Not tested yet	
Xastir https://github.com/Xastir/Xastir	Not tested yet	Digipeating?
Aprx https://github.com/PhirePhy/aprx	Not tested yet	Digipeating support?
Dxspider https://www.dxcluster.org/		
D-rats https://github.com/ham-radio-software/D-Rats/tags	Project archived; low test priority	
LDSPED	No longer required?	
Other programs? Link ideas: https://packet-radio.net/downloads/		
https://github.com/jmkristian/		

- ax25-apps and ax25-tools test results

ax25-apps program:	Test result:
/usr/bin/call /usr/bin/listen	Works (big bin) broken
/usr/sbin/ax25ipd /usr/sbin/ax25mond /usr/sbin/ax25rtctl /usr/sbin/ax25rtd	Not tested Not tested Not tested Not tested

ax25-tools Program #1:	Program #1 test result
/usr/bin/axgetput /usr/bin/mheard /usr/bin/sethdlc /usr/bin/smmixer	Not tested Works For obsolete BayCom ISA card Still useful?
/usr/sbin/ax25_call /usr/sbin/ax25d /usr/sbin/axctl /usr/sbin/axparms /usr/sbin/axspawn /usr/sbin/axwrapper /usr/sbin/beacon /usr/sbin/bppparms /usr/sbin/dmascc_cfg /usr/sbin/kissattach /usr/sbin/kissnetd /usr/sbin/kissparms /usr/sbin/m6pack /usr/sbin/mcs2h /usr/sbin/mheardd /usr/sbin/mkiss /usr/sbin/net2kiss	What is this? Small binary Not tested -- important Not tested -- important Not tested Not tested Not tested Not tested -- important Is this obsolete? For obsolete PI/PI2/PacketTwin cards Obsolete Obsolete? Obsolete Obsolete For obsolete YAM ISA cards Obsoleted by ax25netd Obsolete Obsolete?

ax25-tools program #2:	Program #2 test result:
/usr/sbin/netrom_call /usr/sbin/netromd /usr/sbin/nodesave /usr/sbin/nrattach /usr/sbin/nrparms /usr/sbin/nrsdrv /usr/sbin/rip98d /usr/sbin/rose_call /usr/sbin/rsattach /usr/sbin/rsdwnlnk /usr/sbin/rsmemsiz /usr/sbin/rsparms /usr/sbin/rsuplnk /usr/sbin/rsusers.sh /usr/sbin/rxecho /usr/sbin/tcp_call /usr/sbin/ttylnkd /usr/sbin/yamcfg	What is this? Small binary Not supported yet - important Not supported yet Obsolete Obsolete Obsolete? Not tested What is this? Small binary Not supported yet - ROSE Not supported yet - ROSE Not supported yet - ROSE Not supported yet - ROSE Not supported yet - ROSE Not supported yet - ROSE Obsolete What is this? Small binary Not tested For obsolete YAM ISA cards

Errata

- 2026-10-04: KI6ZHD : ax25netd default AGW tcp port moved from 8100 to 8200 to avoid conflicts
- 2026-10-03: KI6ZHD:
 - Updated socket path to /var/run/ax25/sockets/ax25netd.sock
 - Change /etc/ax25/agwpe.conf to /etc/ax25/ax25netd_agwpe.conf
- 2026-10-01 : KI6ZHD – mentioned listen does not work yet
- 2026-09-30 : KI6ZHD – Updated scenarios; added program compatibility table
- 2026-09-27 : KI6ZHD – various corrections
- 2026-09-26 : KI6ZHD – first version